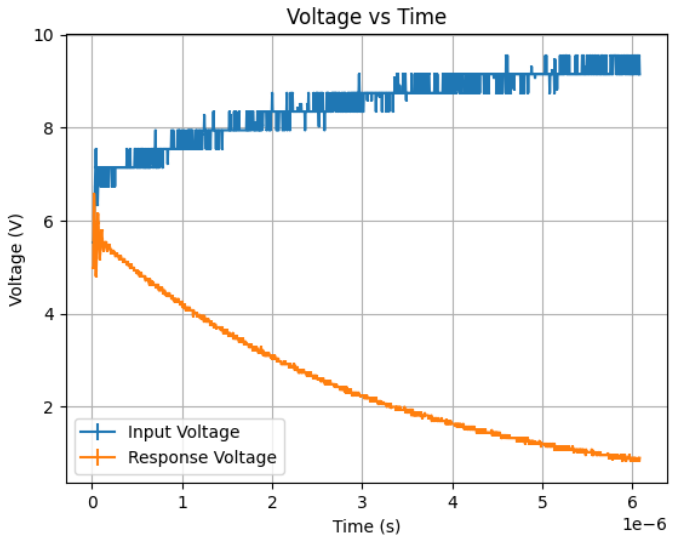
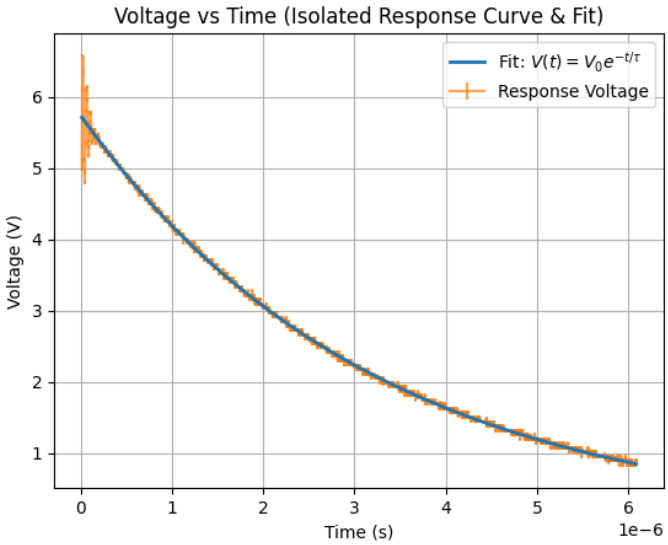
Capacitor Exercise-Report (By Leon Yiliang Qian and Terry Chan)

1. Introduction: This experiment examines how a capacitor charges and discharges in an RC circuit. The time it takes for the voltage across the capacitor to change is described by the time constant $\tau = RC$. We will measure the 90% to 10% fall time and compare it to the theoretical time constant.
2. Experiment method:
    - Connect a resistor and capacitor in series and apply a square wave voltage to the circuit.
    - We chose a 22nF capacitor, and a 100ohm resistor.
    - Use the measure function on the oscilloscope to take measurements of the time it takes for the capacitor voltage to drop from (90%)* to 10% of its maximum value.
    - Use Python to fit the data to the capacitor's theoretical model with the exponential model and compare the results with the manually measured results.
3. Results:
    - Oscilloscope results:



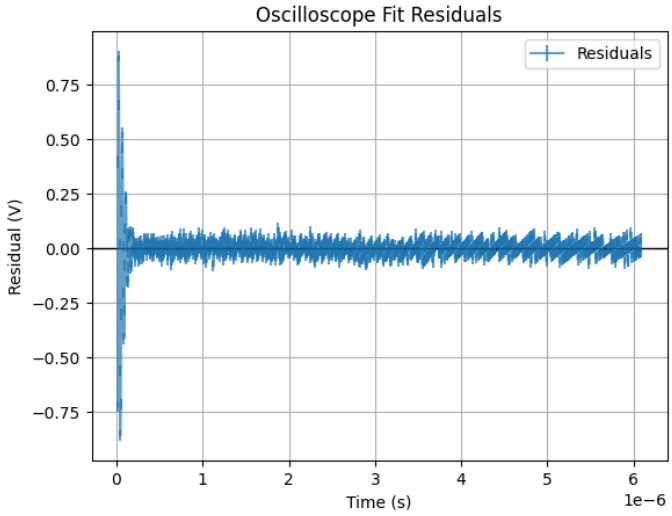
This graph represents the measurements taken from the oscilloscope during the capacitor's charging and discharging cycles. It shows the voltage across the capacitor as it decreases from (75%)* to 10% of its maximum value during discharge relative to time.

*: Initially, we measured the fall time from 90% to 10%, but we noticed that the first part of the graph showed an unexpected curve that didn't match the expected exponential behavior. To address this, we reconfigured the oscilloscope to measure the voltage drop from 75% to 10%.



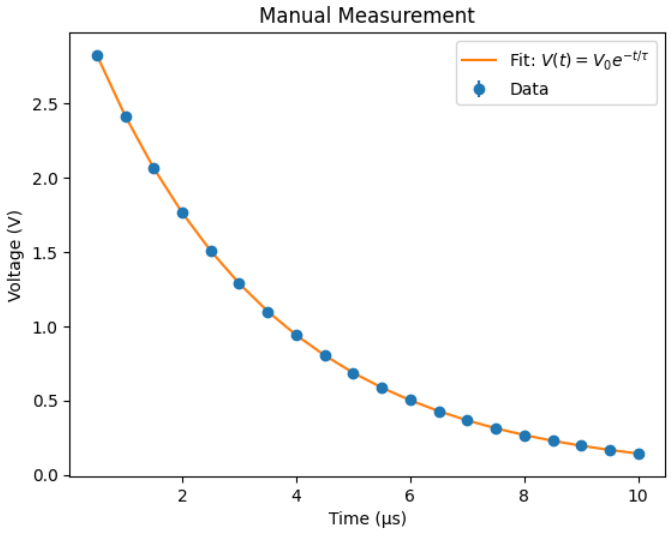
This graph shows the voltage response measured from the oscilloscope as the capacitor discharges over time. The orange curve represents the actual response voltage data, while the blue curve is the best fit of the exponential model $V(t) = V_0 e^{-t/\tau}$.

By printing the data from the curve fit function, we have that $V_0 = 5.7408 \pm 0.0021$ and $\tau = 3.1816 \pm 0.019$, which gives us the fitted model equation: $V(t) = 5.7408e^{-t/3.1816}$, where the time constant has a unit of milliseconds.



This graph shows the residuals between the voltage measured by the oscilloscope and the fitted exponential model over time. The residuals fluctuate randomly around zero, indicating that the model accurately captures the behavior of the capacitor's discharge.

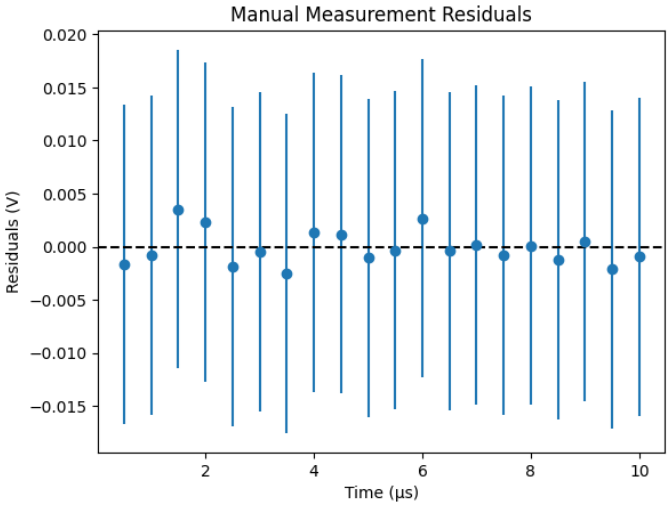
- Manual results:



This graph shows the voltage decay of the capacitor over time as measured manually.

The orange curve represents the best fit line for the exponential model $V(t) = V_0 e^{-t/\tau}$, with the manually meseasured data displayed as the blue dots for comparison. The fit demonstrates the expected exponential behavior of the capacitor discharge.

By printing the data from the curve fit function, we have that $V_0 = 3.310 \pm 0.014$ and $\tau = 3.1775 \pm 0.0183$, which gives us the fitted model equation: $V(t) = 3.310e^{-t/3.1775}$, where the time constant has a unit of milliseconds.



This graph displays the residuals between the manually measured data points and the fitted exponential model over time. The residuals are scattered around zero, with error bars representing measurement uncertainties.

The random distribution of the residuals suggests that the model effectively fits the data, although the larger error bars indicate greater uncertainty in the manual measurements.

- Predicted parameter values:
    - Theoretical time constant: 2.20 μs
    - Time constant as measured by the oscilloscope: 2.73 μs
    - Time constant as calculated from the manual measurement fit: 3.178 $\pm$ 0.018 μs
    - Time constant as calculated from the oscilloscope fit: 3.1816 $\pm$ 0.0019 μs

4. Conclusion: The time constants measured from both the manual and oscilloscope data match closely with the theoretical predictions, confirming that the capacitor's discharge follows the expected exponential behavior. The oscilloscope provided more precise results than our manual measuarements, but both support the theory. (Aside: one interesting thing we noticed was that even when we set 0V in the circuit, the voltmeter sometimes showed a small non-zero reading. This could happen due to leftover charge in the capacitor, noise in the circuit, or interference from nearby components) Overall, the experiment confirms that the time constant $\tau = RC$is accurate, and the model fits the data well.